

MEDB 5501, Module03

2026-09-08

Topics to be covered

- What you will learn
 - The normal distribution
 - Normal probabilities and quantiles
 - Assessing normality
 - Using R to assess normality
 - Your homework

The bell shaped curve

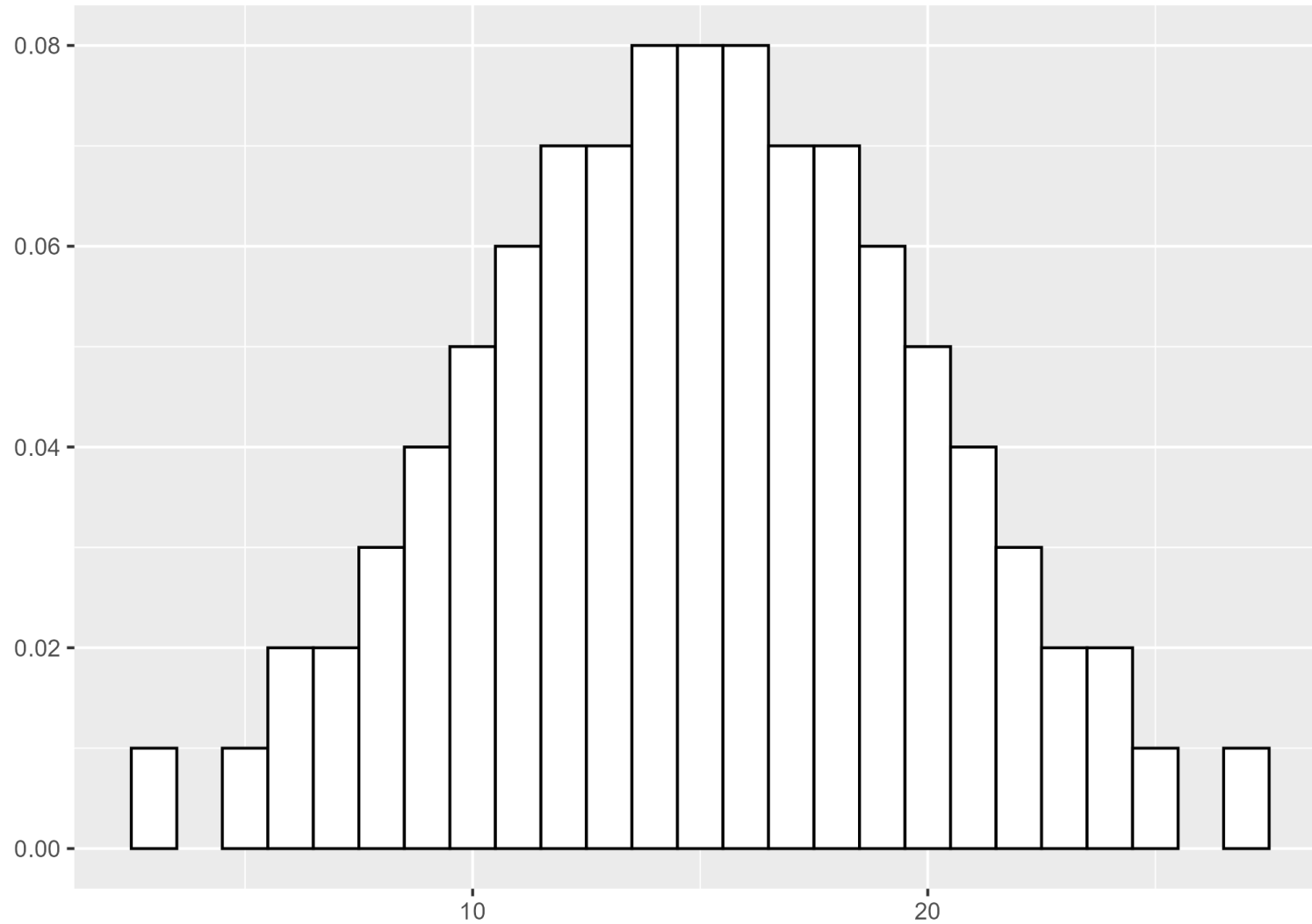
- Does your variation follow a bell shaped curve?
- Synonyms: normality, normal distribution
 - Values in the middle are most common
 - Frequencies taper off away from the center
 - Symmetry on either side

Speaker notes

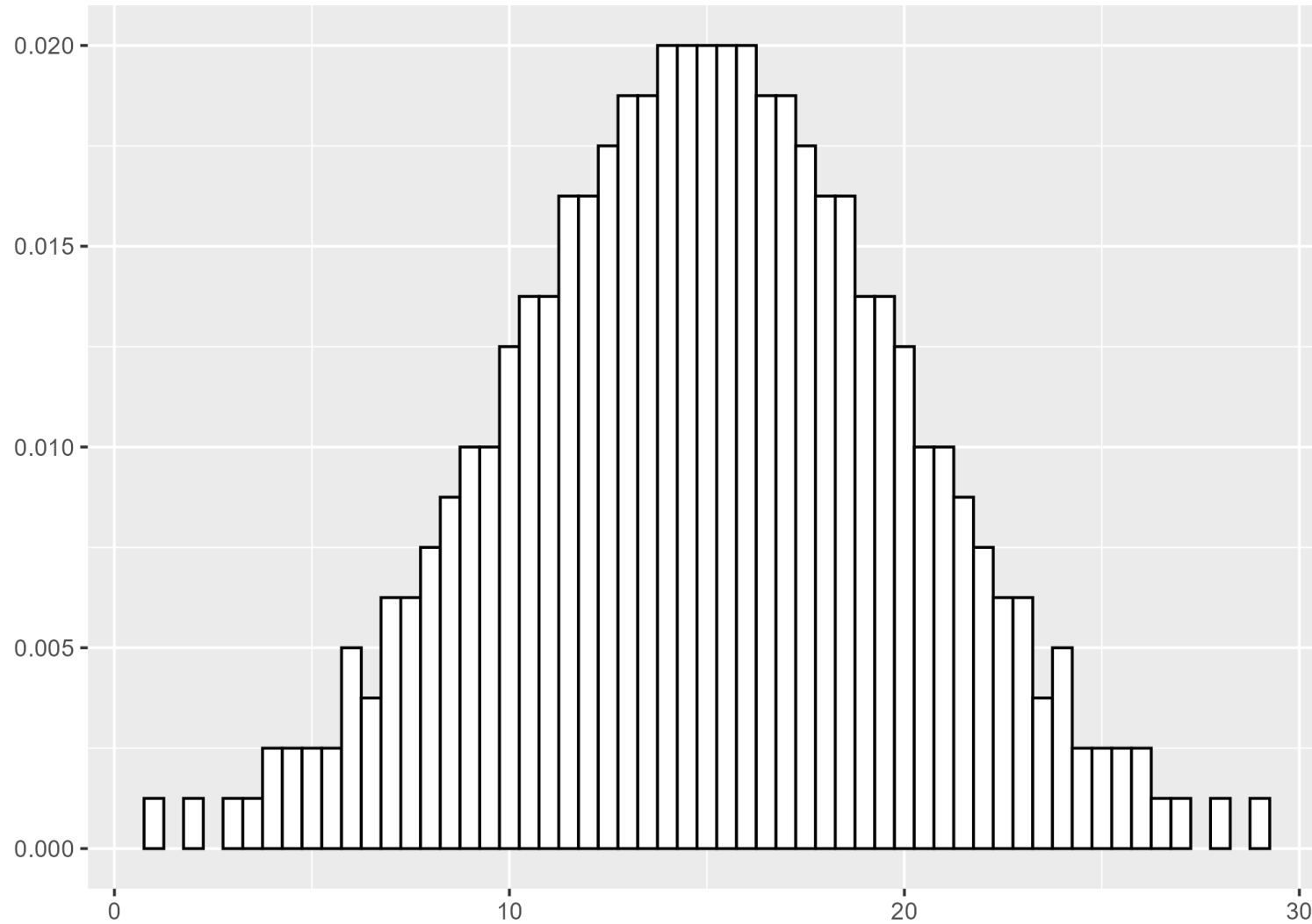
Speaker notes

Much variation in the real world follows a bell shaped curve, alternately called a normal distribution. You can assess whether you have a bell shaped curve using a histogram. Look for values in the middle being most common. The frequencies should taper off slowly as you moved away from the middle. The histogram should have symmetry. The left side of the histogram should be roughly equivalent to the right side of the histogram.

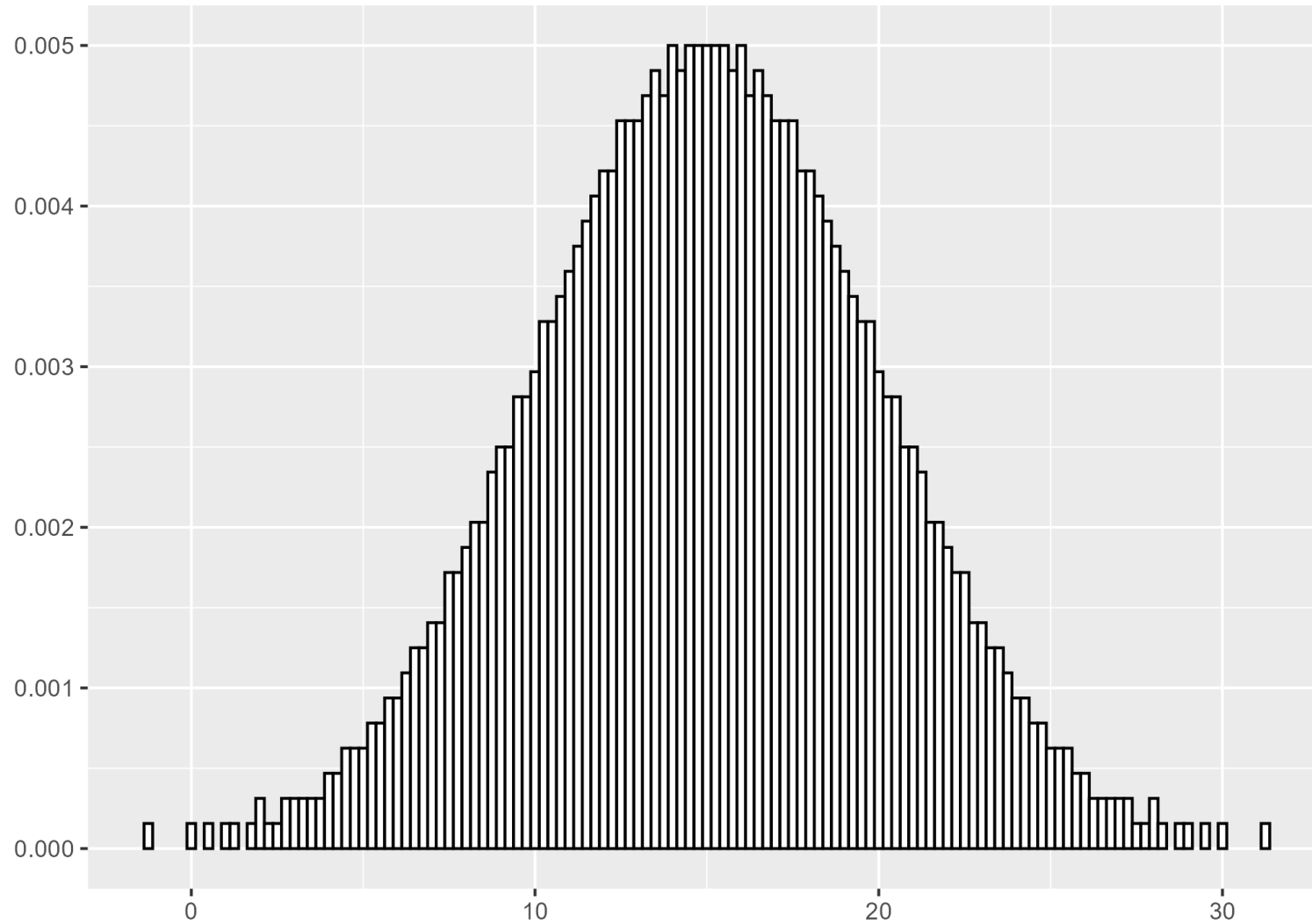
Bell-shaped curve is a limit, 1



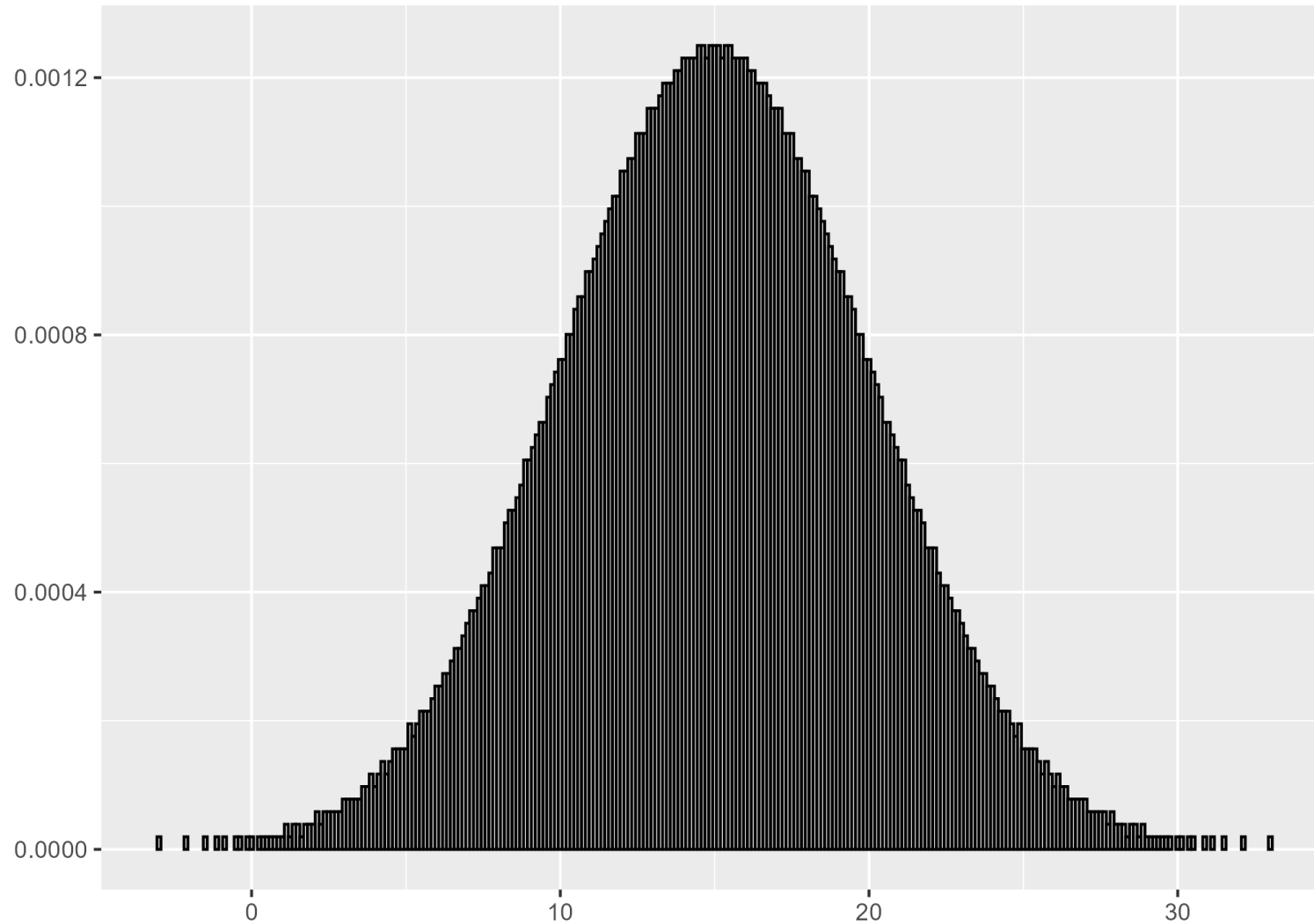
Bell-shaped curve is a limit, 2



Bell-shaped curve is a limit



Bell-shaped curve is a limit



Overview

The normal distribution is an important component of most statistical analyses. The formula for the normal distribution is quite complex,

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

but the shape is readily recognizable.

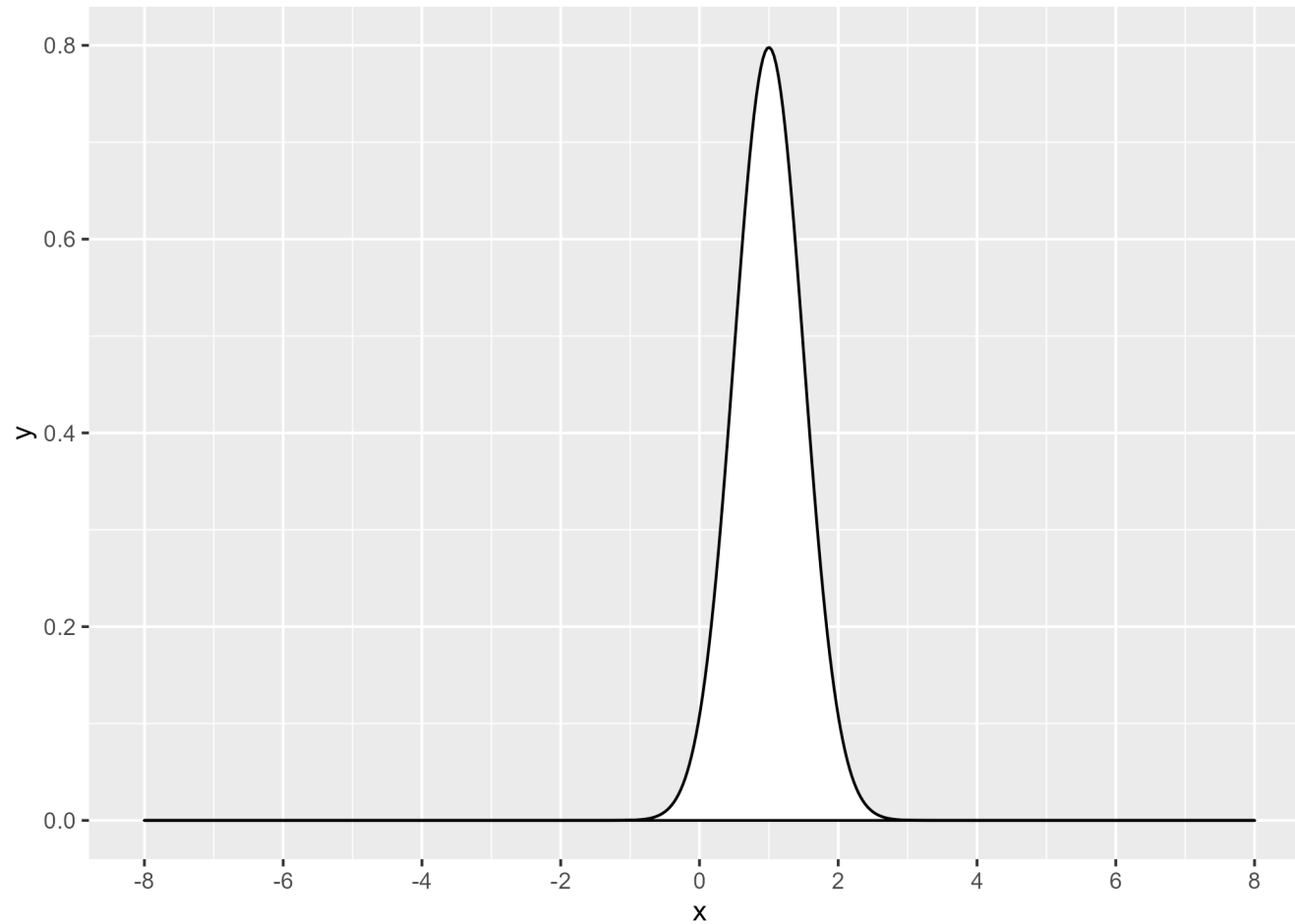
**Normal distribution with $\mu=2$,
 $\sigma=1$**

**Normal distribution with $\mu=-3$,
 $\sigma=1$**

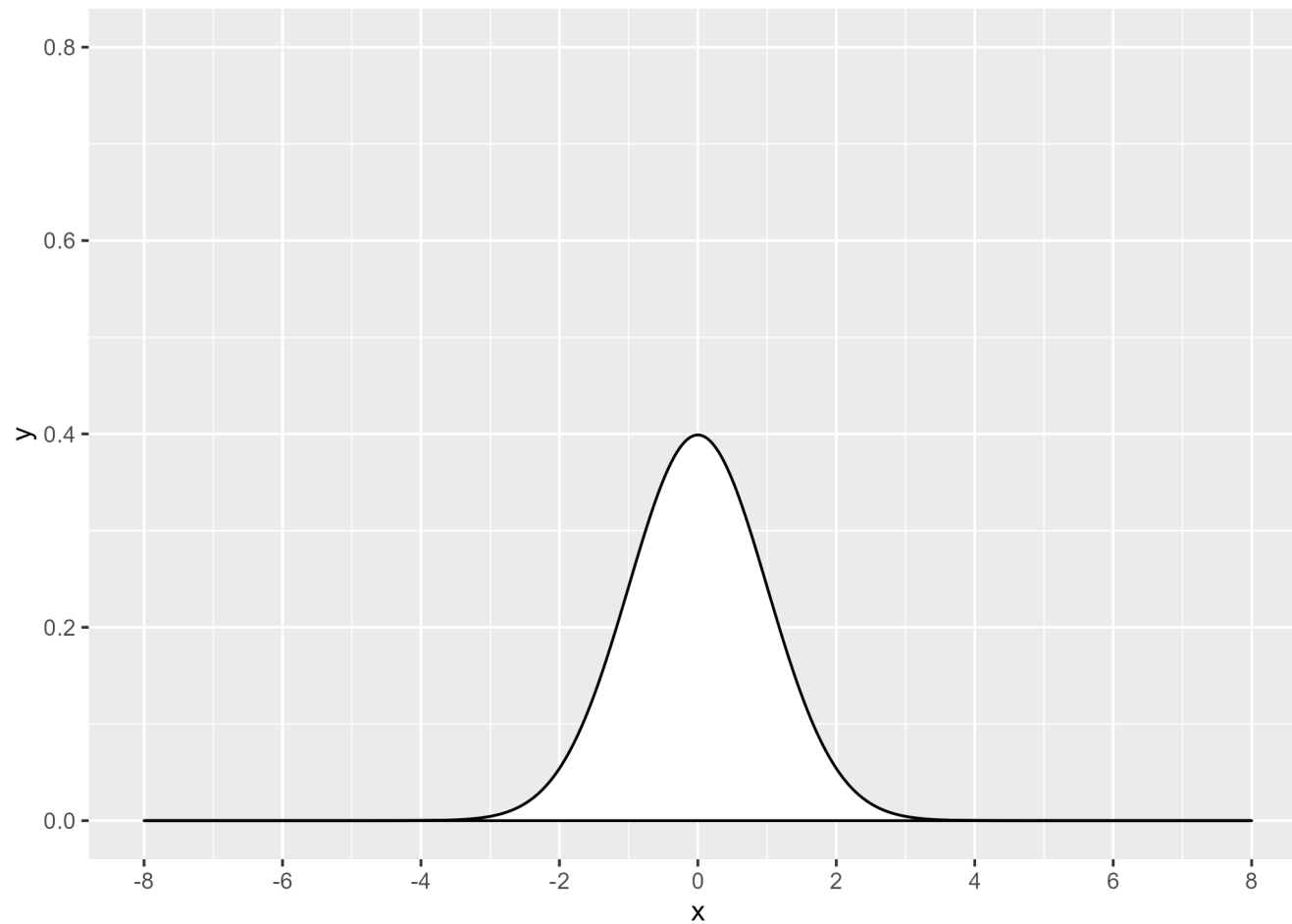
**Normal distribution with $\mu=1$,
 $\sigma=1$**

**Normal distribution with $\mu=1$,
 $\sigma=2$**

Normal distribution with $\mu=1$, $\sigma=0.5$



Standard normal distribution ($\mu=0$, $\sigma=1$)



Why concern yourself with the bell shaped curve?

- You can characterize individual observations
- You can characterize summary measures

Break #1

- What you have learned
 - The normal distribution
- What's coming next
 - Normal probabilities and quantiles

The standard normal distribution, again

Speaker notes

This is the standard normal distribution again. I changed the scaling a bit so it fills up most of the range. I want to show how to do some basic calculations for the standard normal distribution using this graph.

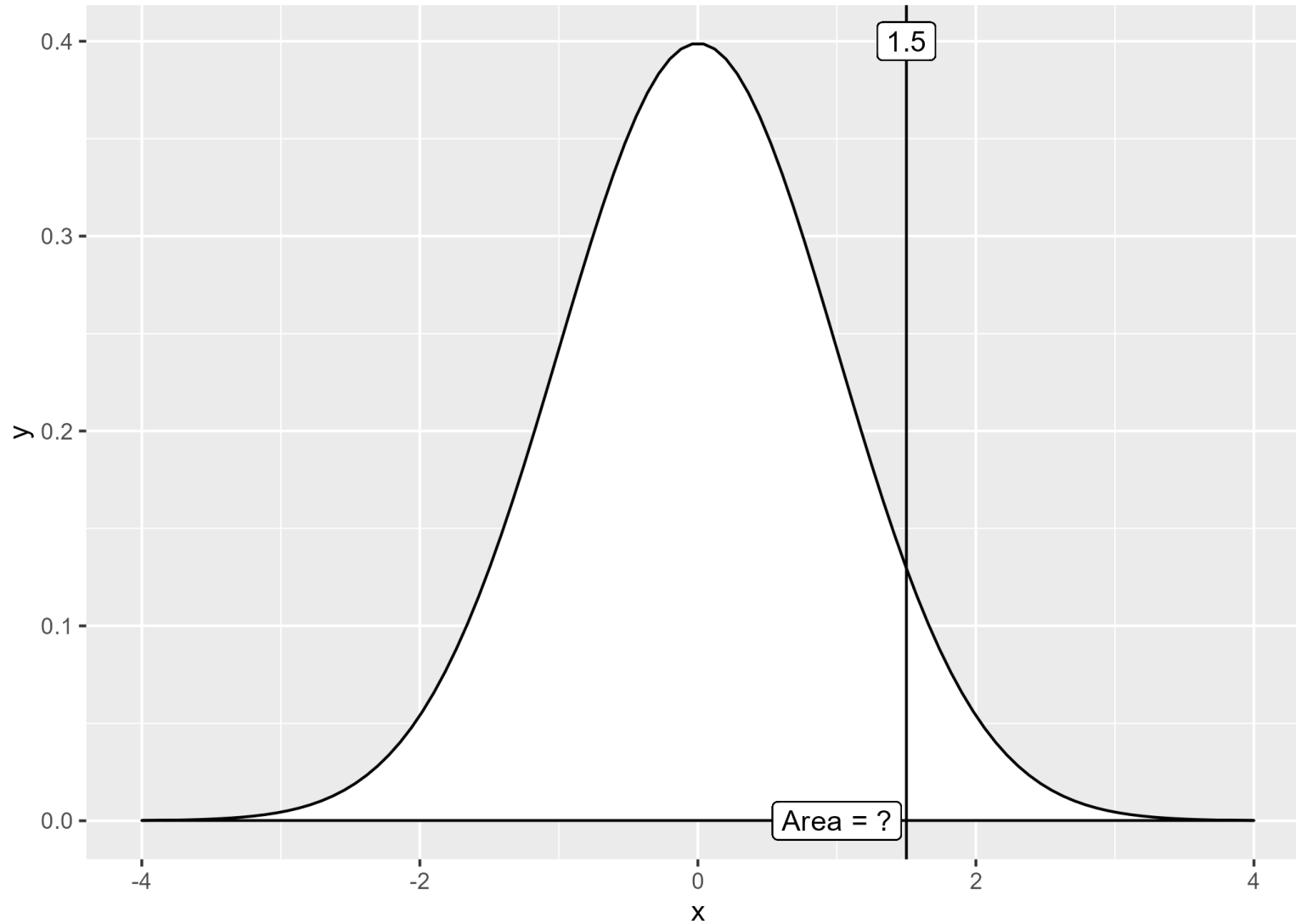
Using calculus to compute probabilities for a normal distribution

$$P[a < X < b] = \int_a^b \left(\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right) dx$$

Speaker notes

For those of you who know calculus, normal probabilities are computed using integration. In simpler terms, probabilities are represented by areas.

Illustrating $P[Z < 1.5]$



Speaker notes

The probability that a standard normal distribution is less than 1.5 is the area of the standard normal curve that lies to the left of 1.5.

Calculating $P[Z < 1.5]$

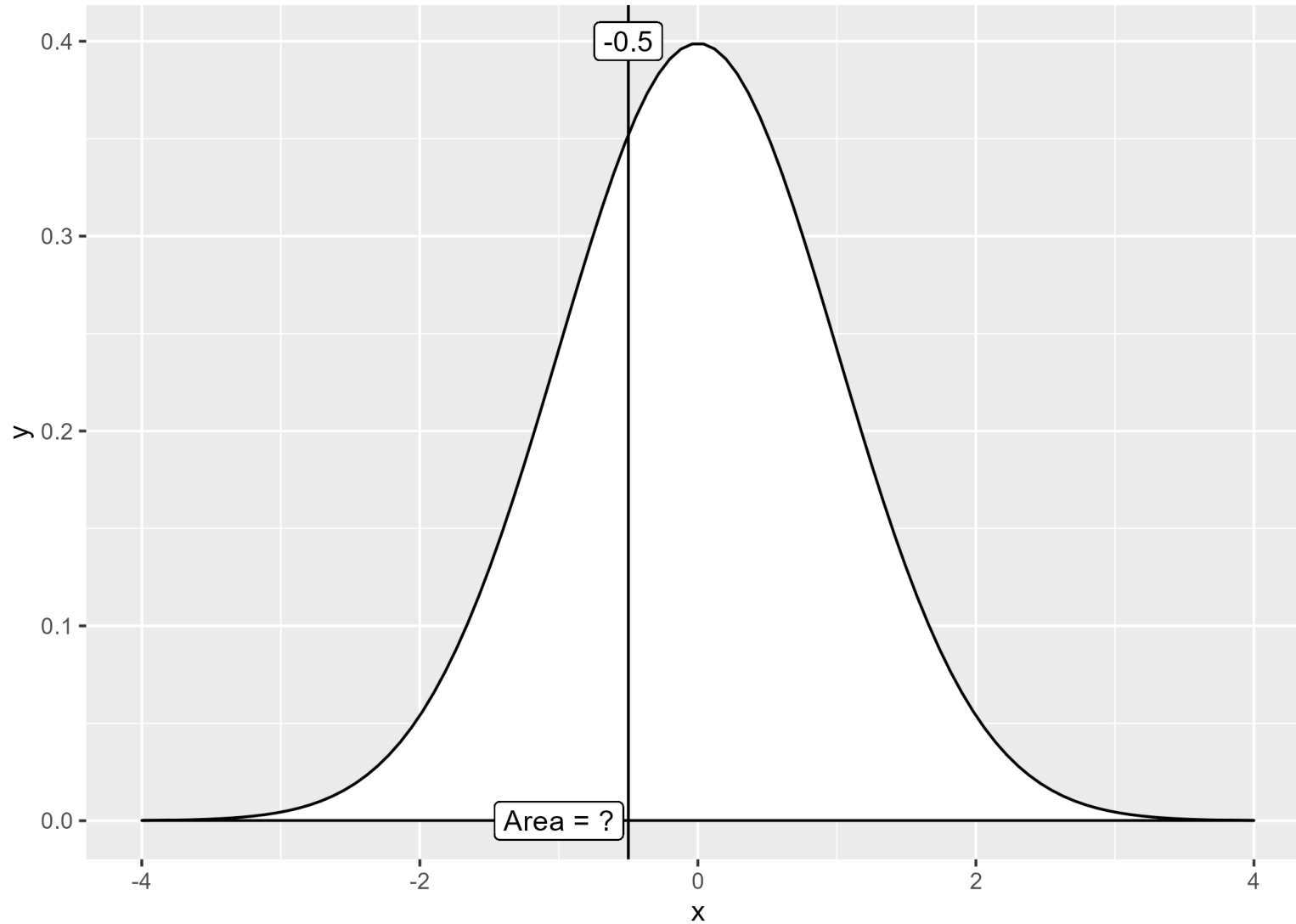
```
1 pnorm(1.5)
```

```
[1] 0.9331928
```


Speaker notes

The probability that a standard normal distribution is less than 1.5 is about 93%.

Illustrating $P[Z < -0.5]$



Speaker notes

The probability that a standard normal distribution is less than -0.5 is the area of the standard normal curve that lies to the left of -0.5 .

Calculating $P[Z < -0.5]$

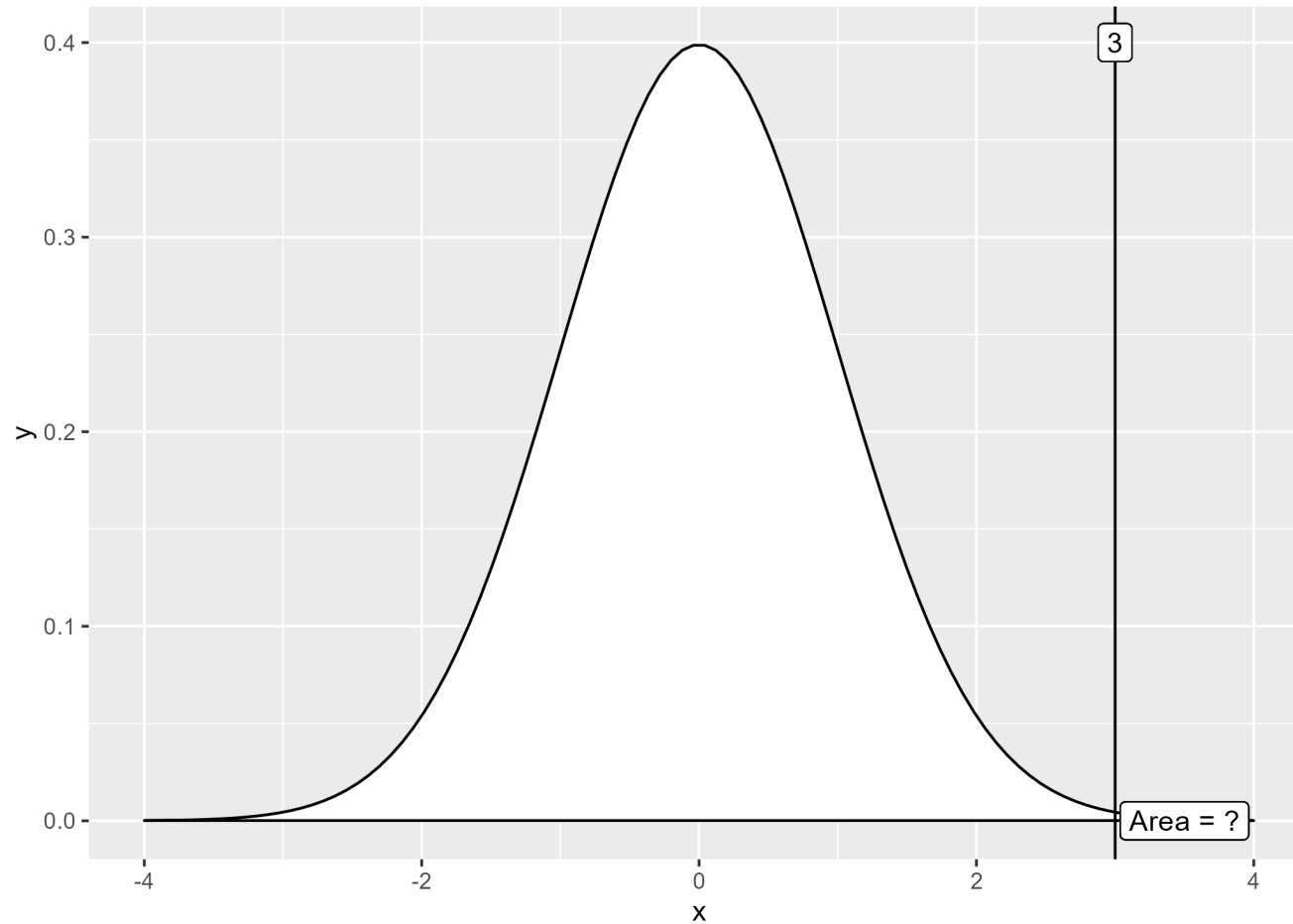
```
1 pnorm(-0.5)
```

```
[1] 0.3085375
```

Speaker notes

Change the argument in the `pnorm` function to get the probability that a standard normal is less than -0.5.

Illustrating $P[Z > 3]$



```
1 1-pnorm(3)
```

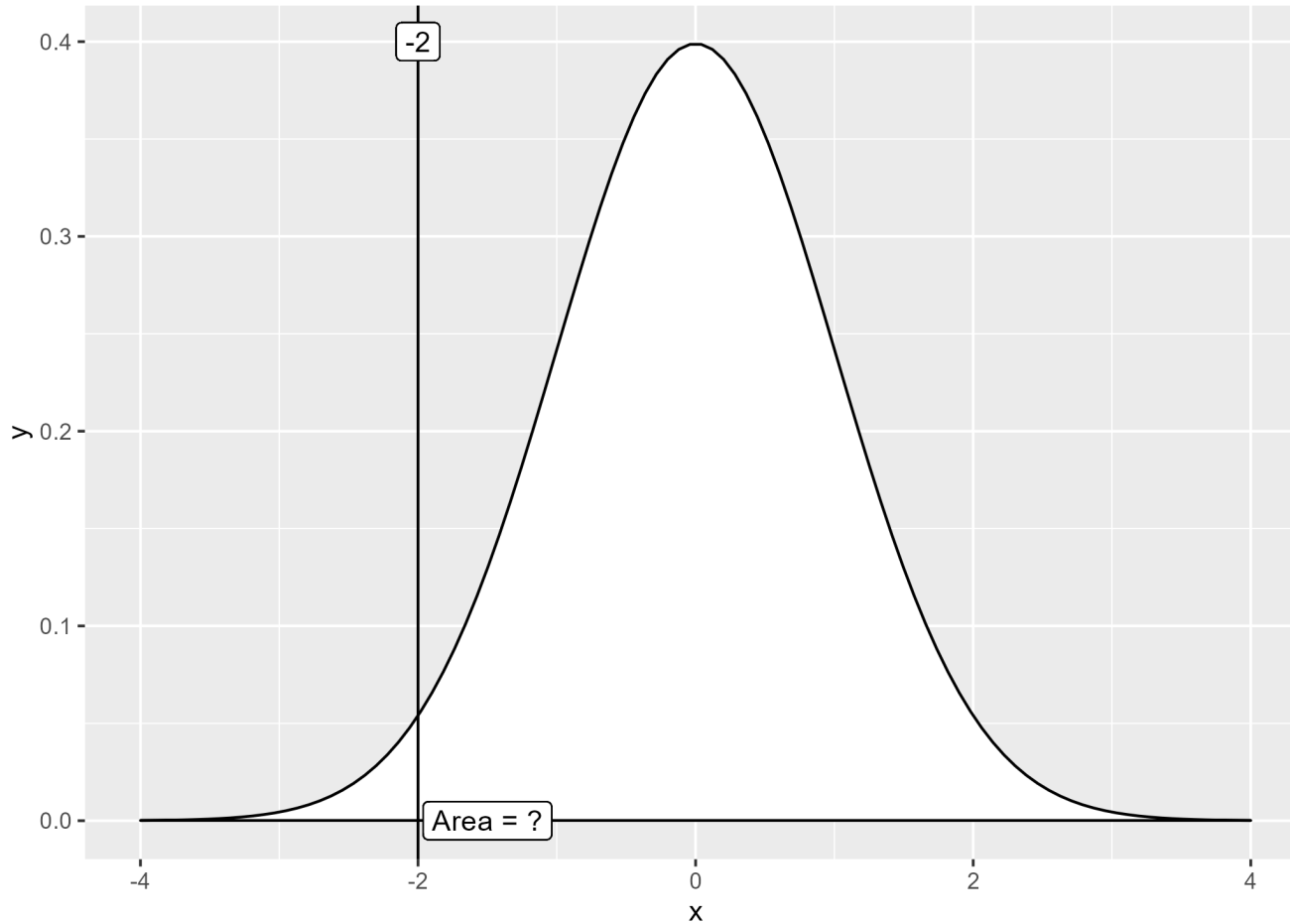
```
[1] 0.001349898
```

Speaker notes

To find a probability of being greater than a specific value, you need to compute the area to the right. This image shows what area corresponds to the probability that a standard normal is greater than 3.

The area to the right of 1 plus the area to the left of 1 has to equal the total probability. So the area to the right is $1 - \text{the area to the left}$. The probability that a standard normal distribution is greater than 3 is about 0.1%.

Illustrating $P[Z > -2]$



```
1 1-pnorm(-2)
```

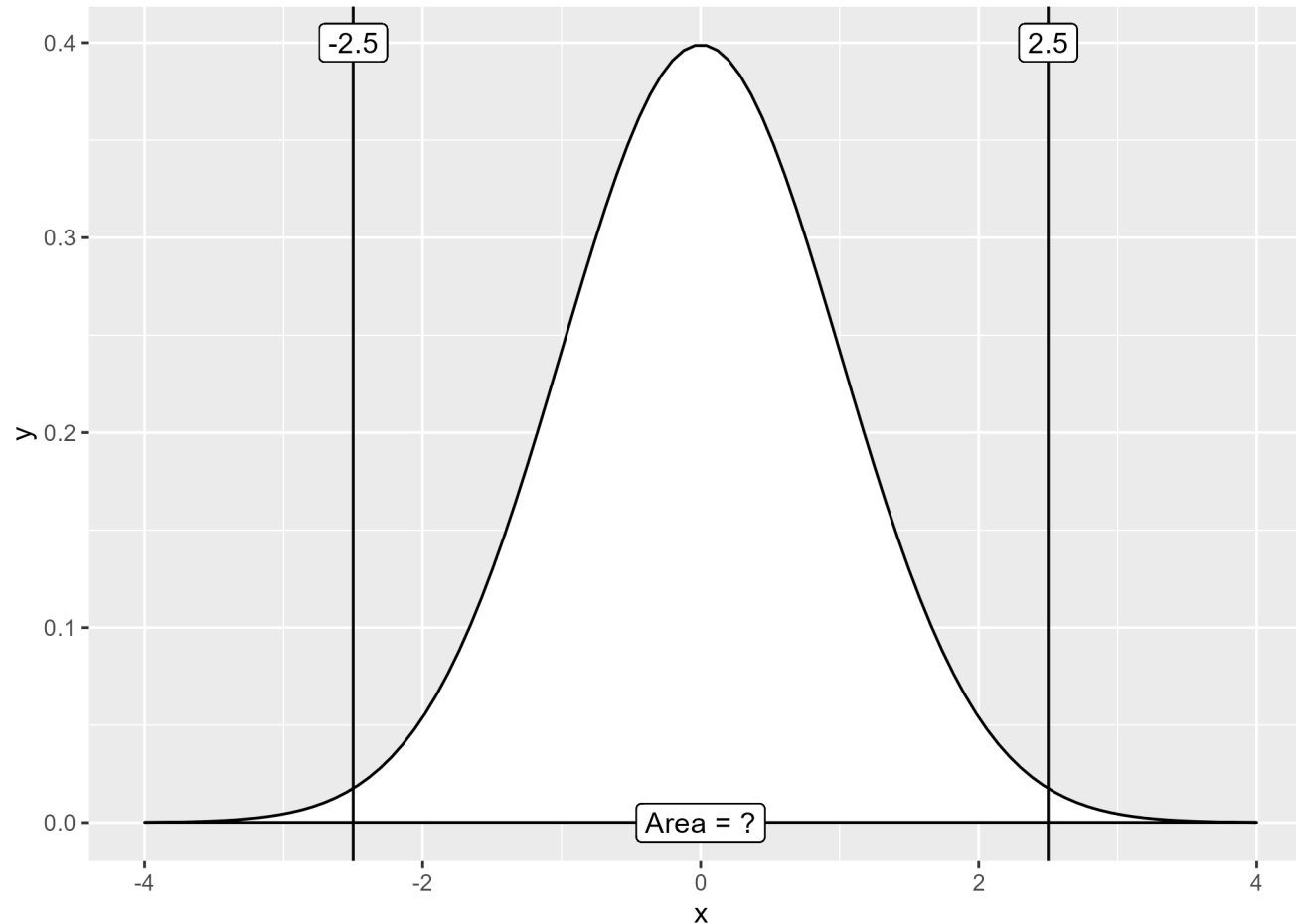
```
[1] 0.9772499
```


Speaker notes

Similarly, the probability for the right of -2 plus the area to the left equals 1. So just subtract the probability to the left from one.

You calculate the area to the right of -2, subtract $\text{pnorm}(-2)$ from 1. The probability that a standard normal is greater than -2 is about 98%.

Illustrating $P[-2.5 < Z < 2.5]$



```
1 pnorm(2.5)-pnorm(-2.5)
```

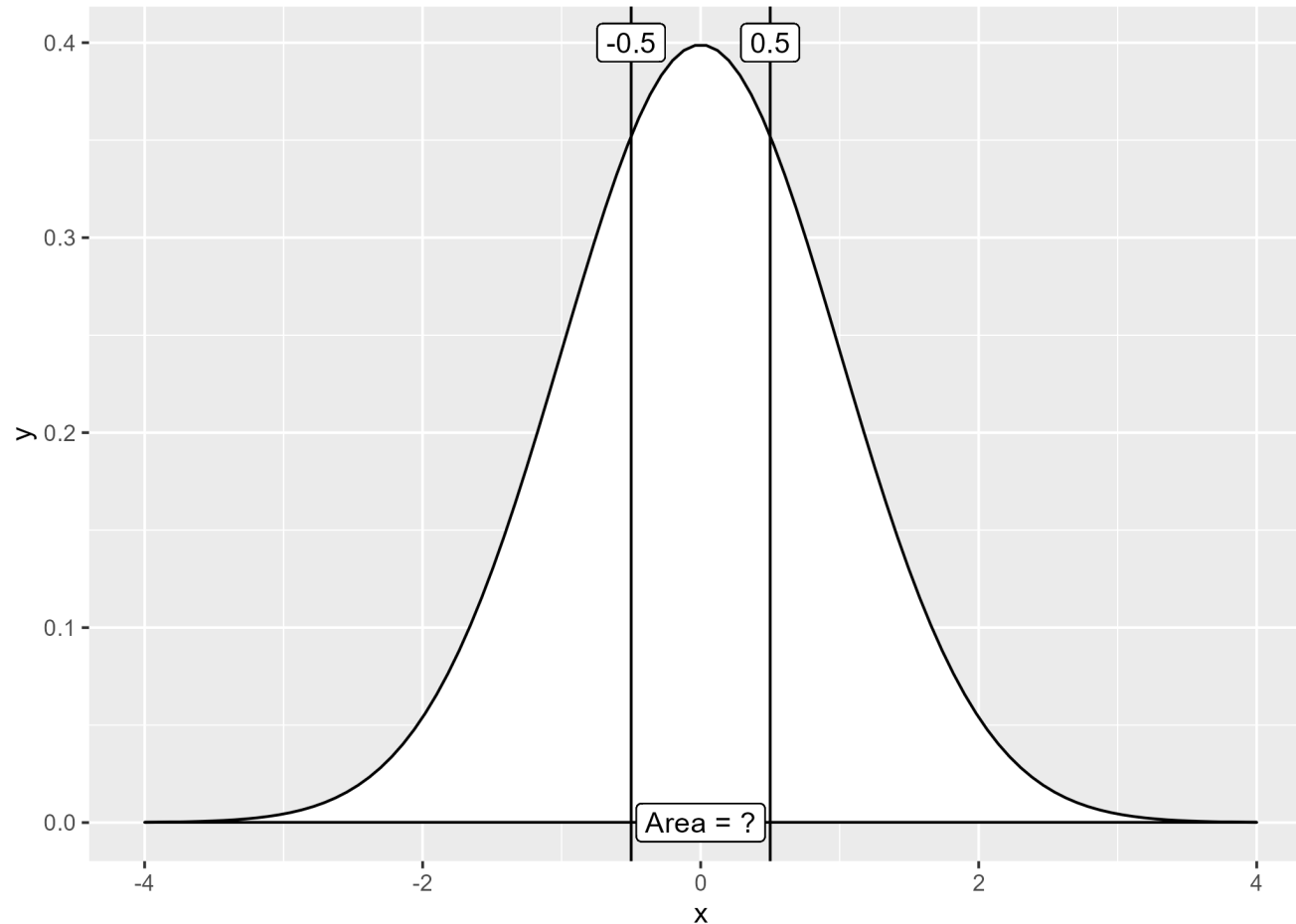
```
[1] 0.9875807
```

Speaker notes

The area between -2.5 and 2.5 is the area to the left of 2.5 minus the area to the left of -2.5.

To get the probability that a standard normal falls between two number, calculate the difference between two pnorm results. The probability that a standard normal is between -2.5 and 2.5 is about 99%.

Illustrating $P[-0.5 < Z < 0.5]$



```
1 pnorm(0.5)-pnorm(-0.5)
```

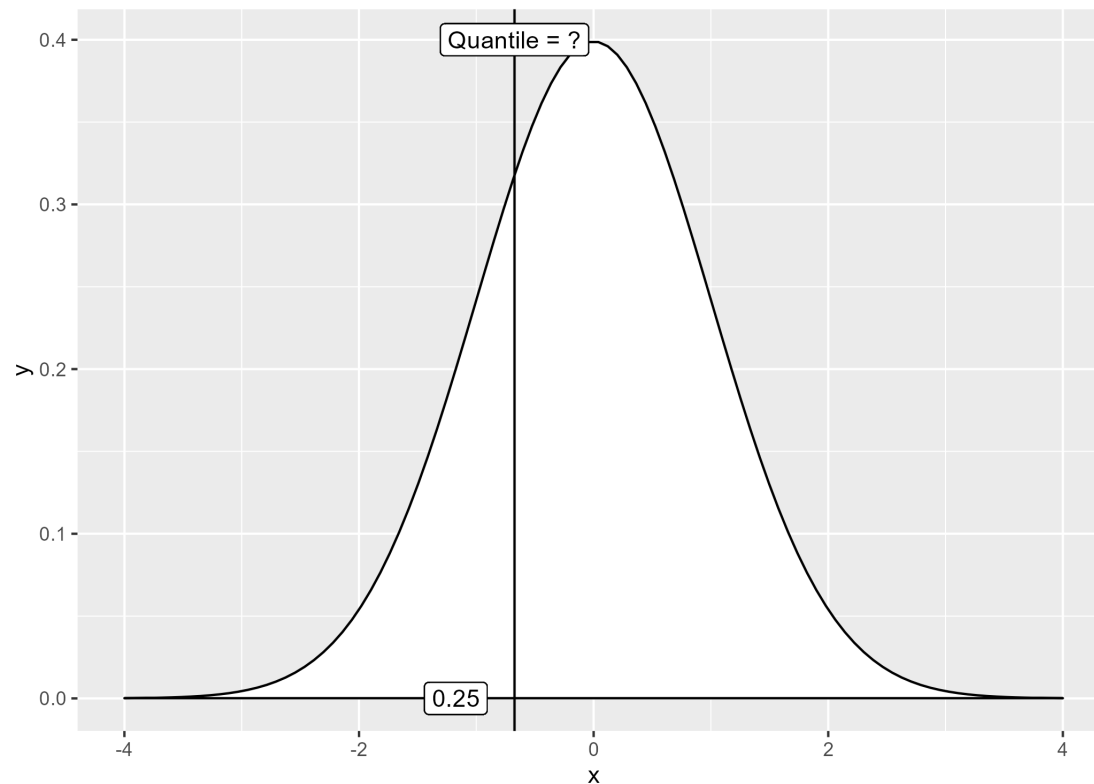
```
[1] 0.3829249
```

Speaker notes

The area between -2.5 and 2.5 is the area to the left of 2.5 minus the area to the left of -2.5.

To get the probability that a standard normal falls between two number, calculate the difference between two pnorm results. The probability that a standard normal falls between -0.5 and 0.5 is about 38%.

Illustrating the 25th percentile of a standard normal



```
1 qnorm(0.25)
```

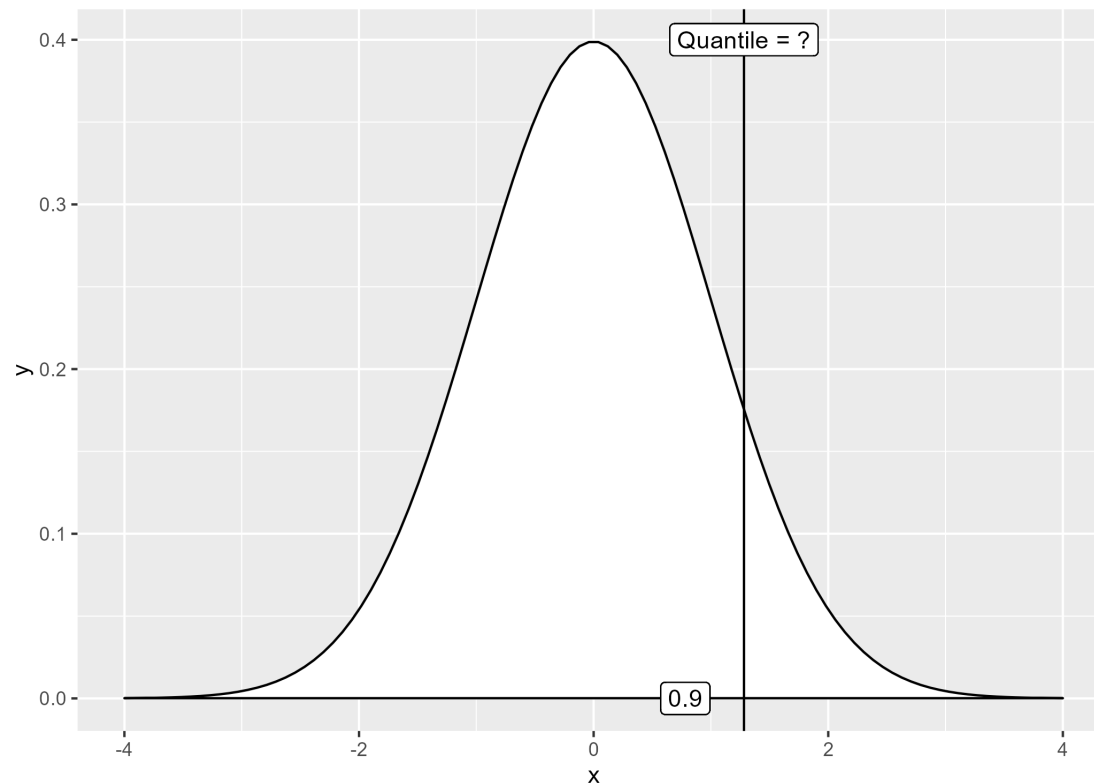
```
[1] -0.6744898
```

Speaker notes

The 25th percentile is the value such that 25% of the area is to the left and 75% is to the right.

To get the percentile, use the `qnorm` function. The 25th percentile of a standard normal distribution is -0.67.

Illustrating the 90th percentile of a standard normal



```
1 qnorm(0.9)
```

```
[1] 1.281552
```


Speaker notes

The 90th percentile is the value such that 90% of the area is to the left and 10% is to the right.

Compute the 90th percentile using `qnorm(0.9)`. The 90th percentile of a standard normal distribution is 1.28.

Probabilities and percentiles for other normal distributions

- If X is $N(\mu, \sigma)$, then
 - $P[X < a] = P[Z < \frac{a-\mu}{\sigma}]$
 - $X_p = \mu + Z_p\sigma$

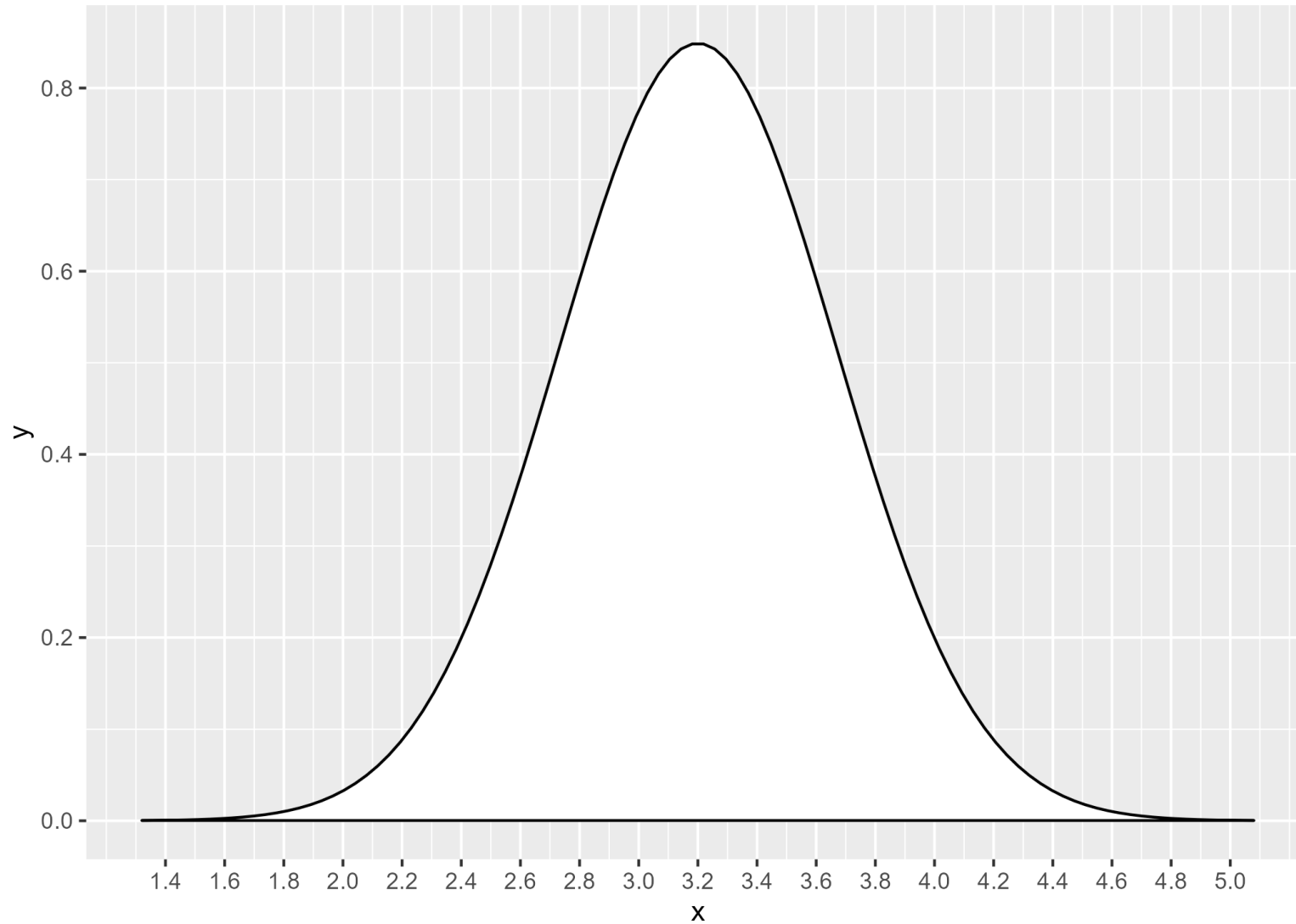
Speaker notes

If you have a normal distribution with a mean other than 0 and a standard deviation other than 1, the probability that this normal distribution is less than a value “a” is the probability that a standard normal is less than $(a - \mu) / \sigma$. Subtracting μ will move shift the value of a to be relative to zero instead of relative to μ . Dividing by σ will stretch or squeeze the value of a to match a standard deviation of 1 instead of a standard deviation of σ .

To get the pth percentile of a normal distribution with mean μ and standard deviation σ , multiply the pth percentile of a standard normal by σ and add μ . Multiplying by the standard deviation will stretch or squeeze the bell shaped curve, depending on whether the standard deviation is larger than or smaller than 1. Adding the mean shifts the bell shaped curve to the left or right, depending on whether the mean is less than zero or greater than zero.

R allows you to automate these transformations with the mean and sd arguments.

Birthweight (kg) is $N(,)$



Probability birthweight > 4 kg

```
1 1-pnorm(4, mean=3.2, sd=0.5)
```

```
[1] 0.05479929
```

Speaker notes

The probability of a big baby, more than 4 kilograms is about 5%.

10th percentile for birthweight

```
1 qnorm(0.1, mean=4.2, sd=0.5)
```

```
[1] 3.559224
```

Speaker notes

The 10th percentile for birthweight is 3.6 kilograms.

Break #2

- What you have learned
 - Normal probabilities and quantiles
- What's coming next
 - Assessing normality

Assessing normality

- No variable follows a perfect normal distribution
 - But many are close
- How to assess (approximate) normality
 - Histogram
 - Boxplot
 - Normal probability plot
- Avoid formal tests of normality

Speaker notes

While no variables are going to fit the bell shaped curve exactly, many variables come close. Your subjective interpretation of graphs is the best way to assess normality, recognizing that you would be satisfied with a reasonable approximation to normality. Three graphs used commonly for this are the histogram, the boxplot and the normal probability plot.

Histogram

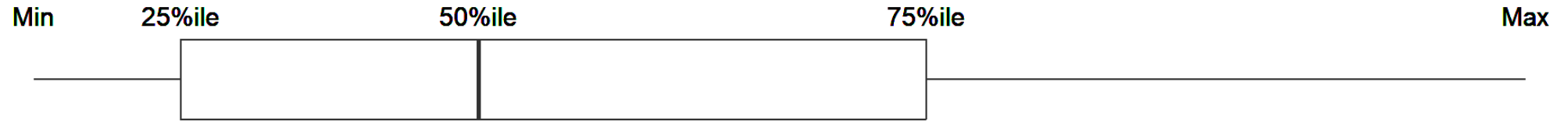
- Peak in the middle
- Roughly symmetric
- Falls off exponentially
- Warning!! Bar width can influence your interpretation
 - Try two or more bar widths

Speaker notes

Look for a peak in the middle of the histogram. If you see two separate peaks, the data is not normally distributed. The histogram should be roughly symmetric, but don't expect perfect symmetry. The bars should fall off more or less exponentially from either side of the peak.

Be careful, because the number of bars that you draw can influence your interpretation. Something that looks normal with a small number of wide bars might look not so normal with a large number of narrow bars.

Sample Boxplot



What to look for in the boxplot

- Median halfway between 25th and 75th percentile.
- Whiskers are same size
- Whiskers not too short, not too long

Constructing a normal probability plot, 1

- Calculate rank
- Divide by $(n+1)$
- Compute corresponding normal percentiles
- Compare these on a graph to the original data
- Roughly straight line implies normality

Constructing a normal probability plot, 2

Use the rank function to assign 1 to the smallest value, 2 to the next smallest value, etc. up to n for the largest value.

	x	r
1	7	4
2	3	2
3	23	9
4	2	1
5	5	3
6	13	6
7	11	5
8	17	7
9	19	8

Constructing a normal probability plot, 3

Divide the rank by (n+1) to get evenly spaced percentages.

	x	r	pctile
1	7	4	0.4
2	3	2	0.2
3	23	9	0.9
4	2	1	0.1
5	5	3	0.3
6	13	6	0.6
7	11	5	0.5
8	17	7	0.7
9	19	8	0.8

Constructing a normal probability plot, 4

Calculate percentiles from a normal distribution using the evenly spaced percentages.

	x	r	pctile	z
1	7	4	0.4	-0.2533471
2	3	2	0.2	-0.8416212
3	23	9	0.9	1.2815516
4	2	1	0.1	-1.2815516
5	5	3	0.3	-0.5244005
6	13	6	0.6	0.2533471
7	11	5	0.5	0.0000000
8	17	7	0.7	0.5244005
9	19	8	0.8	0.8416212

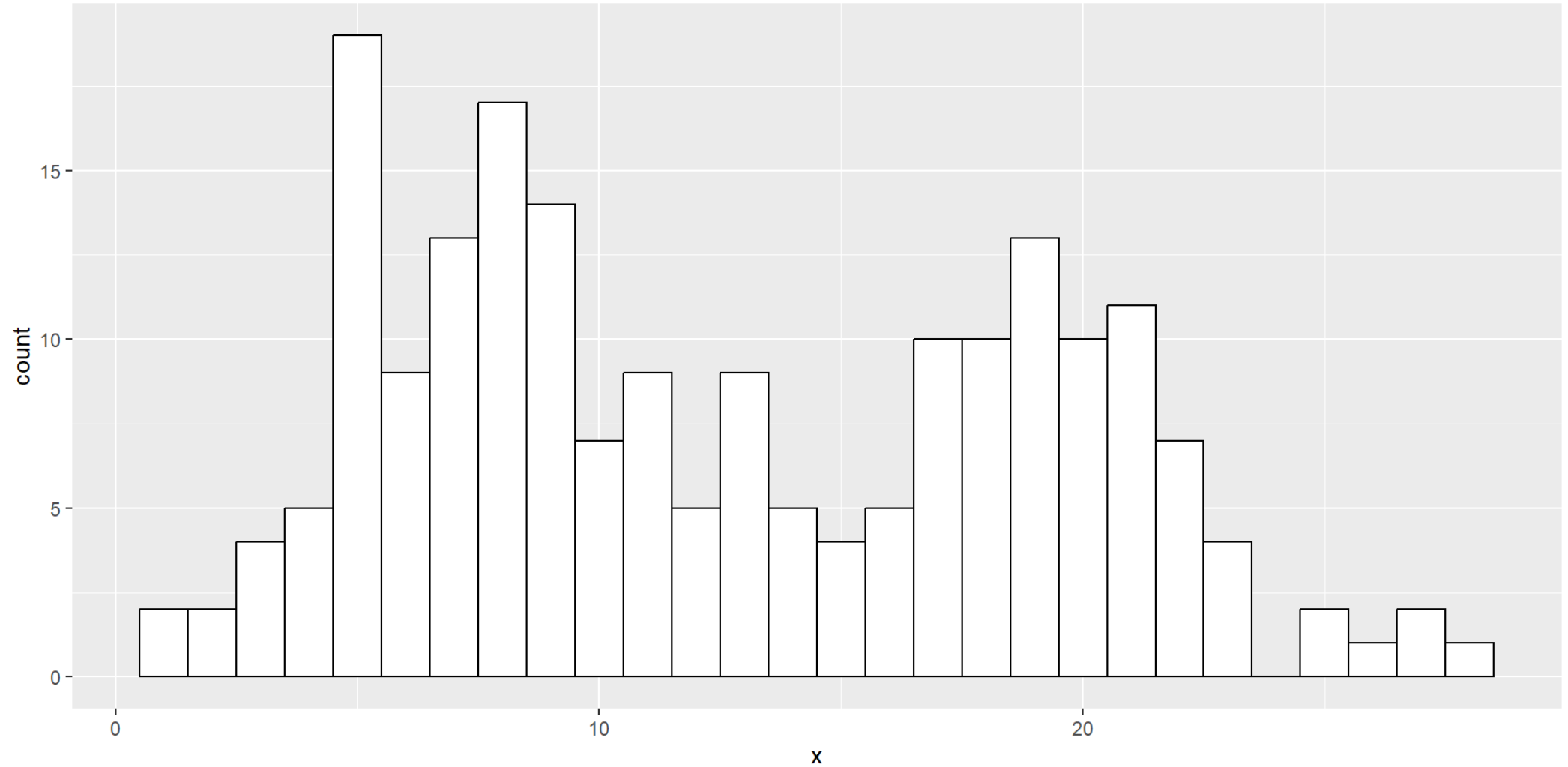
Constructing a normal probability plot, 5

Plot the percentiles from the normal distribution to the original data. A reasonably straight line is evidence of normality.

Constructing a normal probability plot, 6

The `qqnorm` function will do all these steps for you automatically.

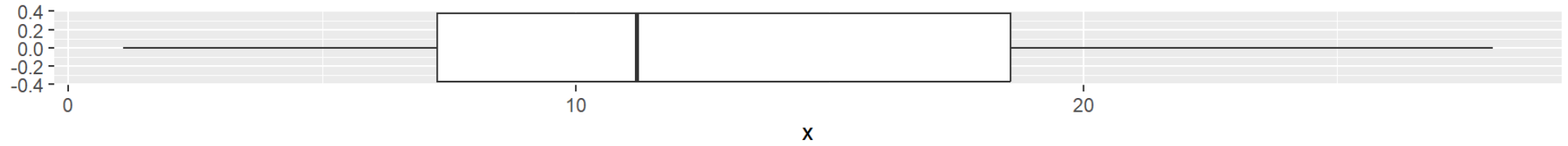
Bimodal data, histogram



Speaker notes

Here's a histogram that shows a bimodal distribution. The frequencies are not highest in the center of the data. This is not a bell shaped curve.

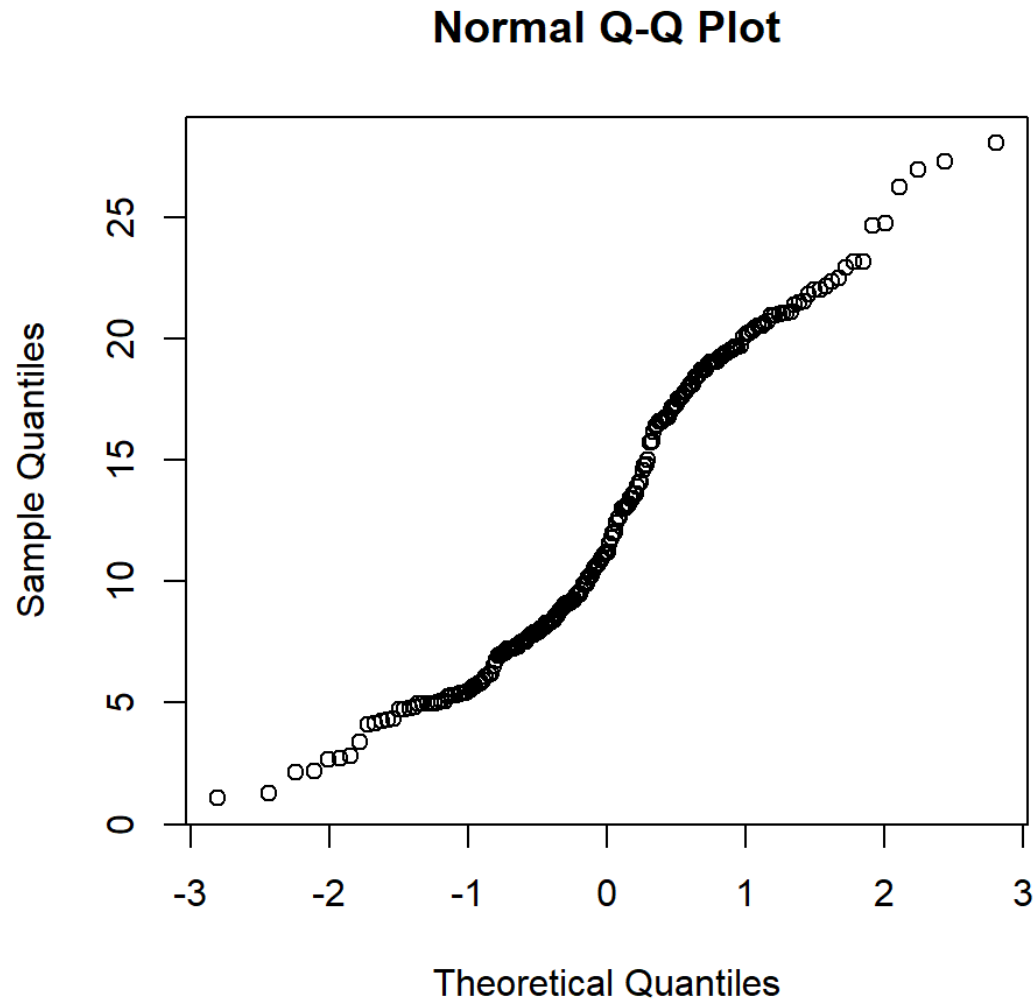
Bimodal data, boxplot



Speaker notes

The boxplot is not as useful as the histogram for detecting bimodal distributions.

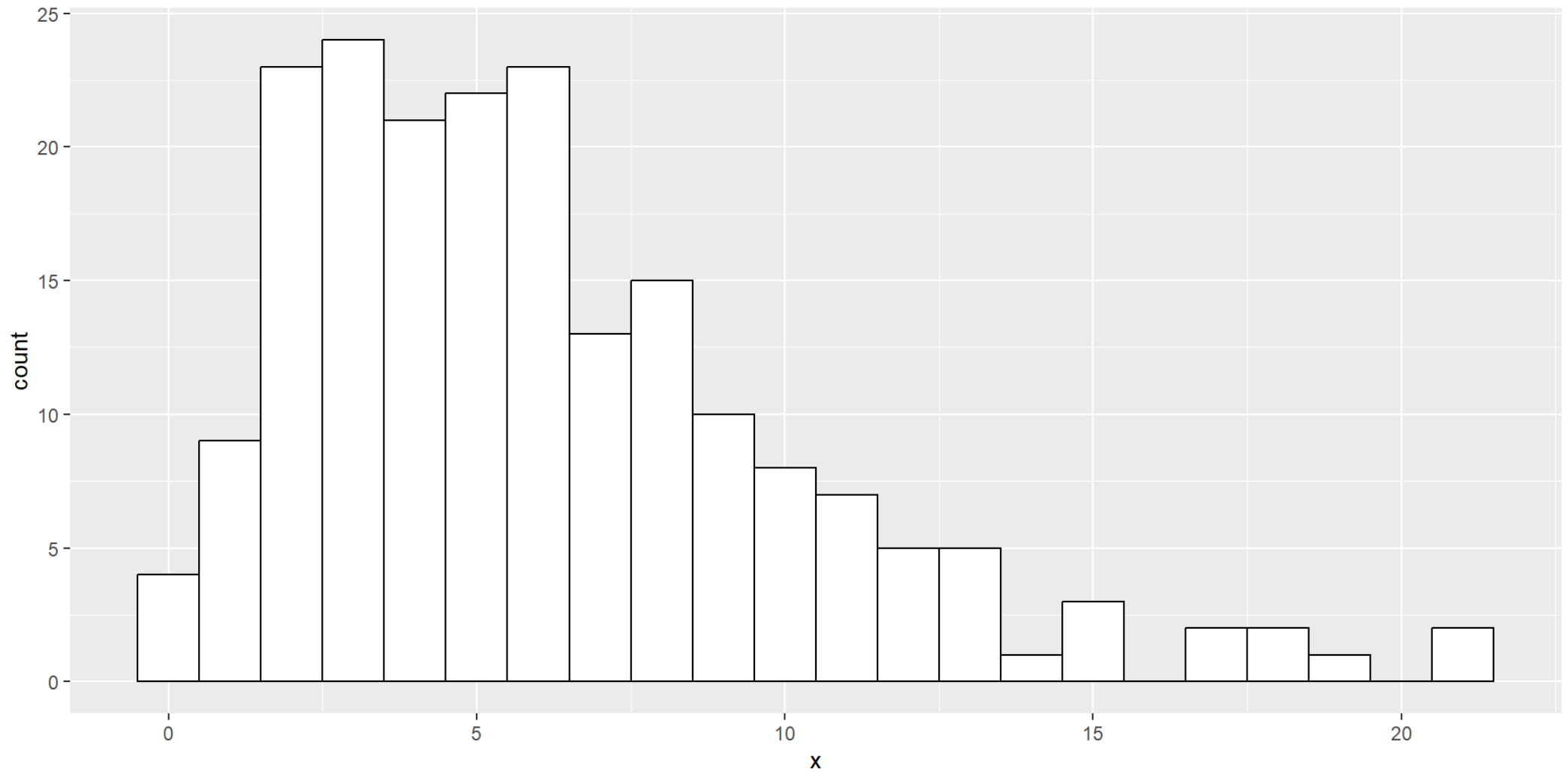
Bimodal data, qq plot



Speaker notes

On the qq plot, a bimodal pattern is often represented as two lines with a sharp jump between them.

Skewed data, histogram

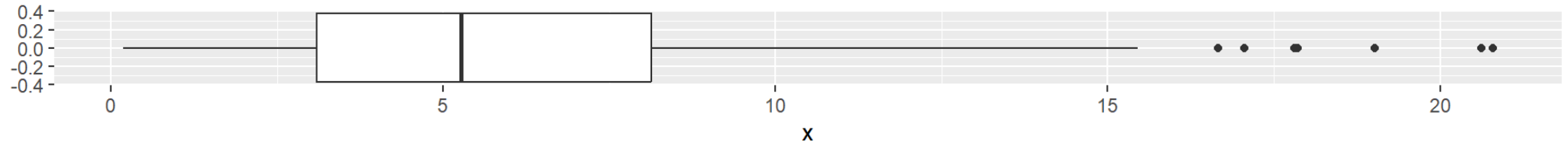


Speaker notes

Speaker notes

Here's a histogram that shows a skewed or asymmetric distribution. This is not a bell shaped curve.

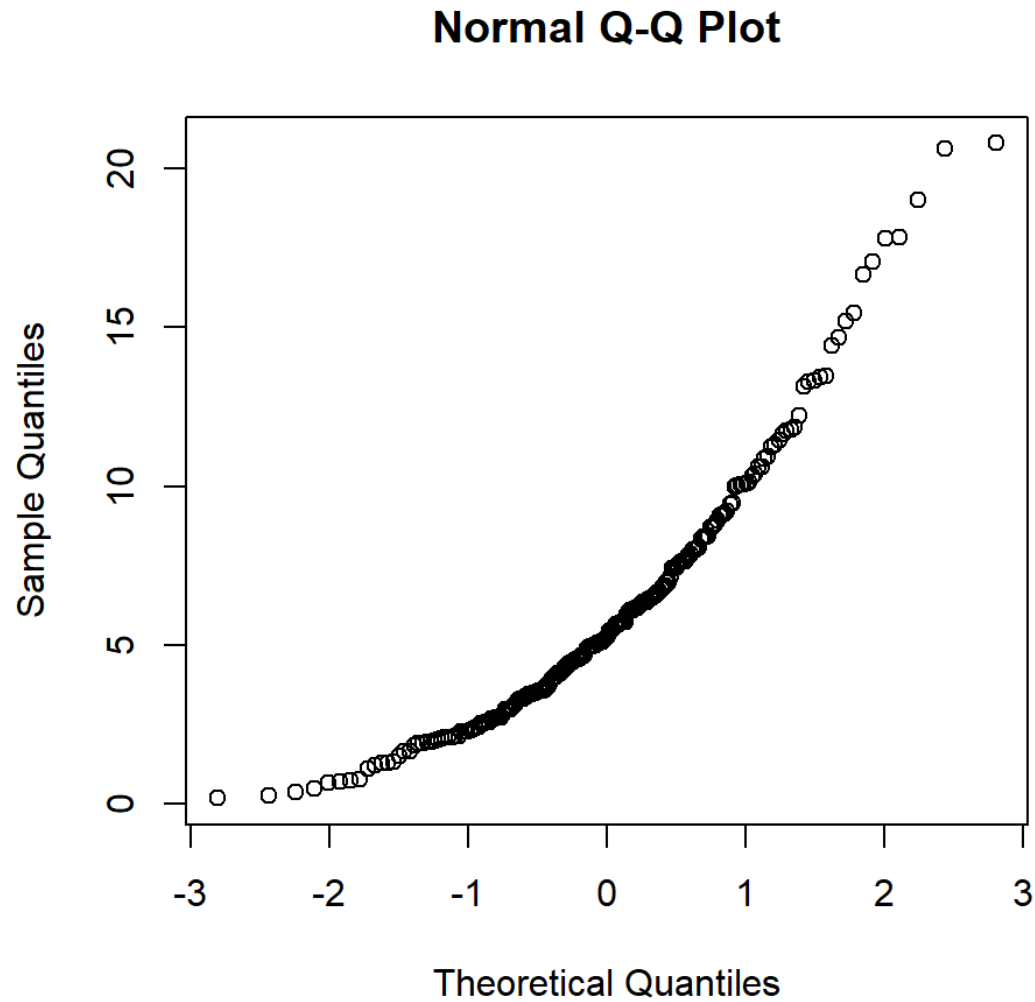
Skewed distribution, boxplot



Speaker notes

An asymmetry in the box and/or the whiskers is an indication of a skewed distribution.

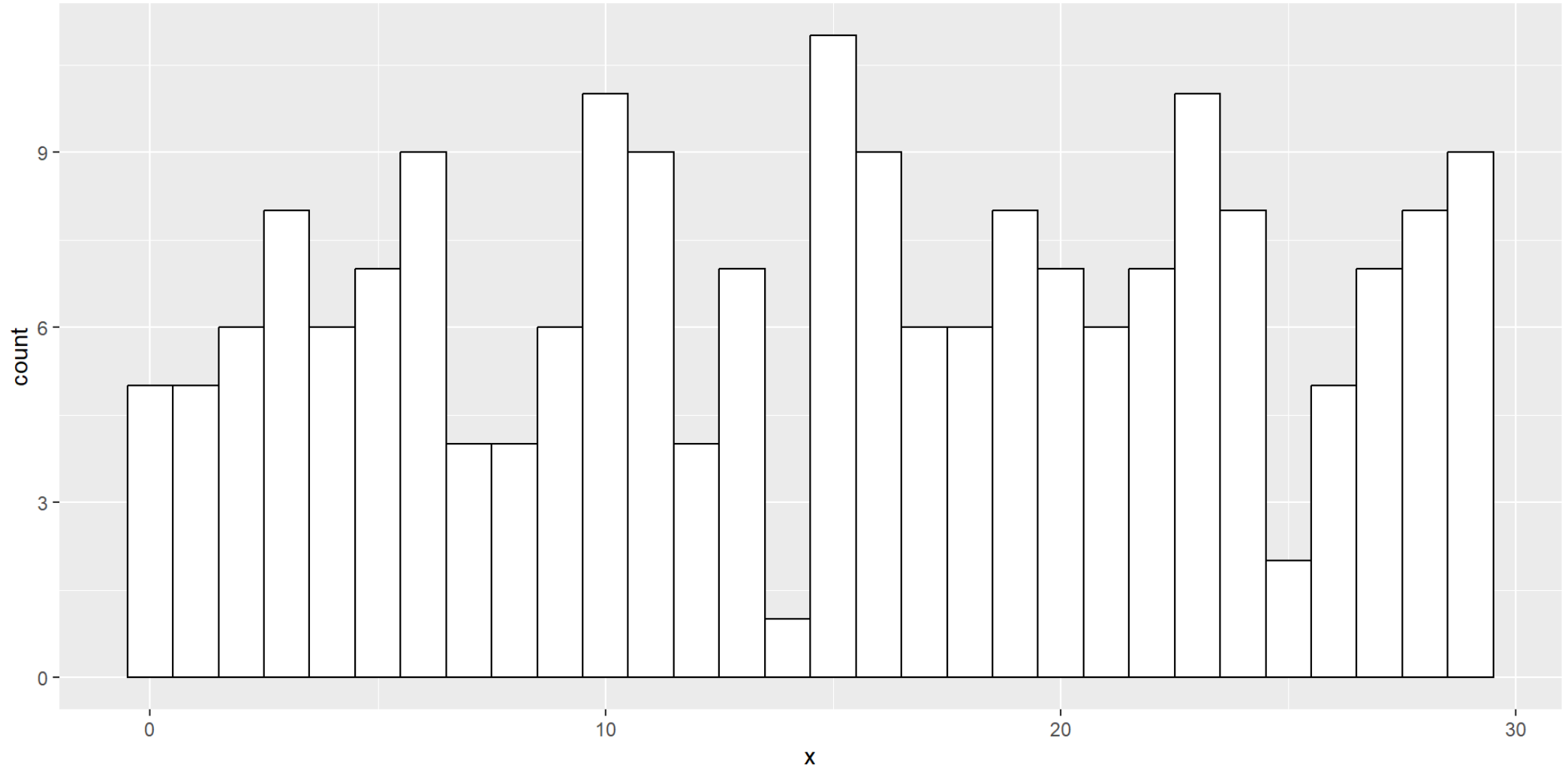
Skewed distribution, qq plot



Speaker notes

A curved pattern for the normal probability plot indicates skewness.

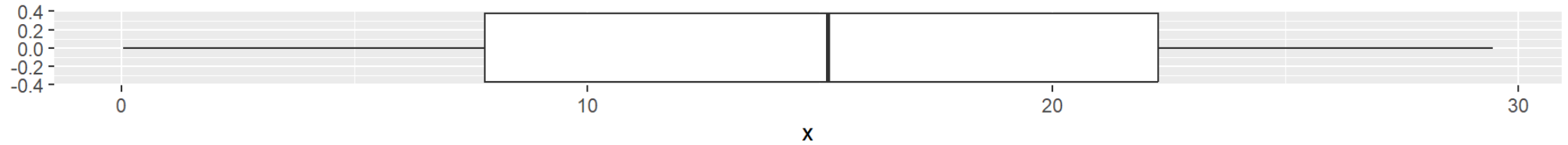
Light-tailed data, histogram



Speaker notes

Here's a histogram that shows a symmetric distribution, but the frequencies do not taper off as you move away from the center. This is not a bell shaped curve.

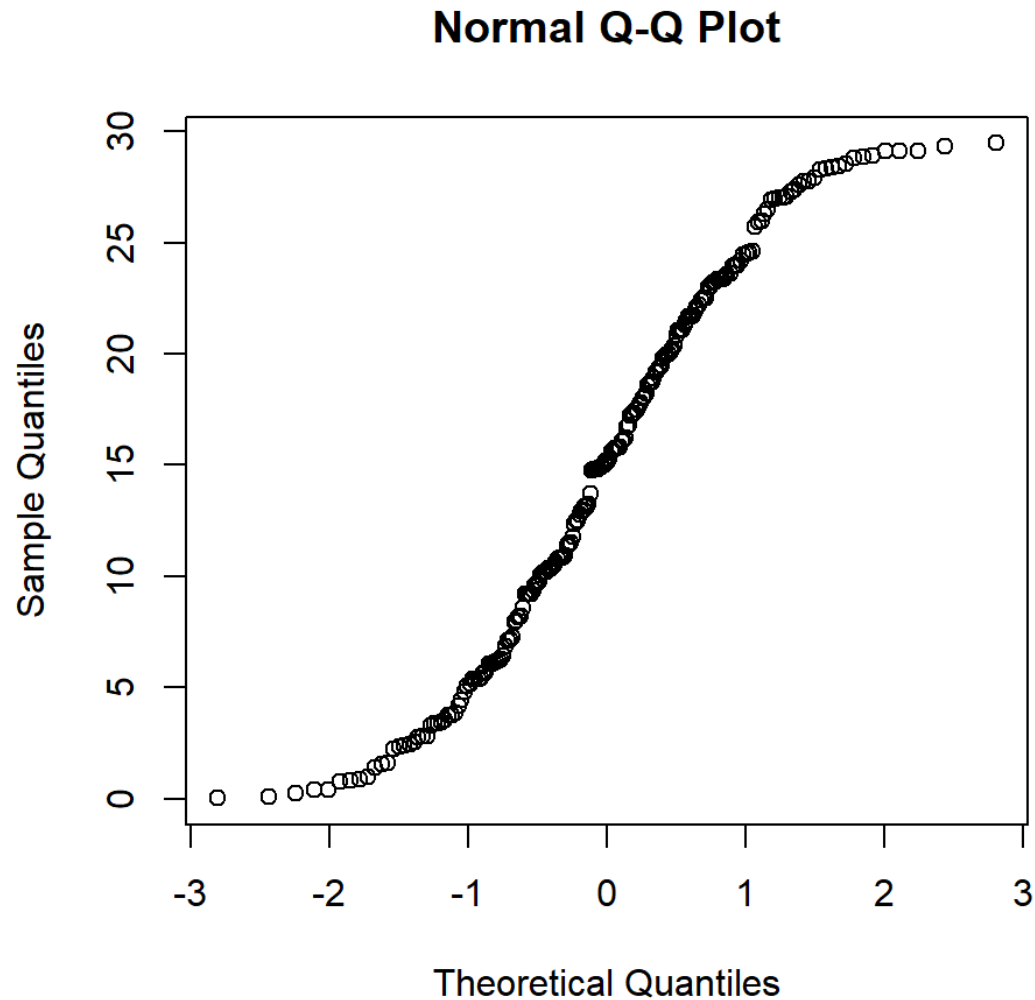
Light-tailed distribution, boxplot



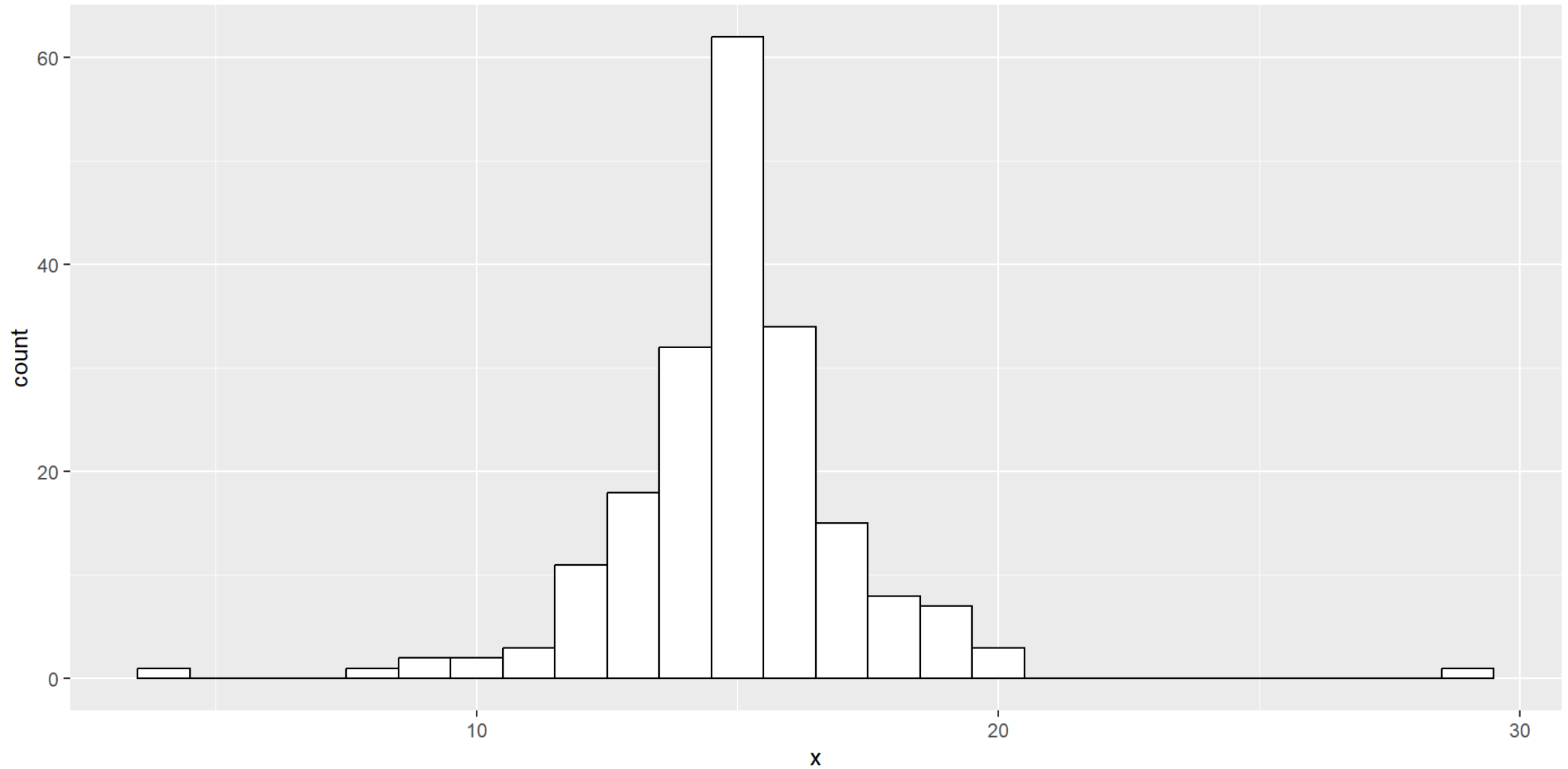
Speaker notes

A boxplot with very short whiskers is evidence of a light tailed distribution.

Light-tailed distribution, qq plot



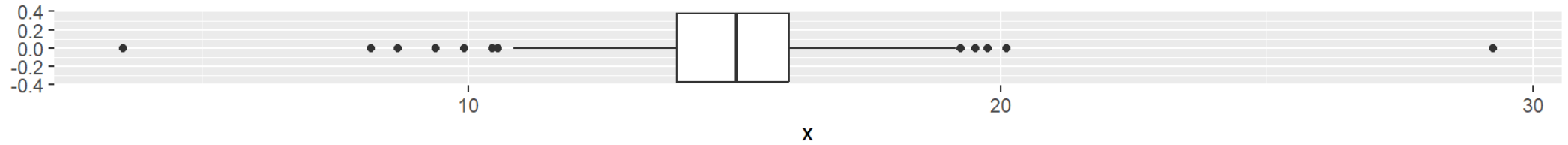
Heavy-tailed distribution, histogram



Speaker notes

Here's a histogram that shows a symmetric distribution, but the frequencies taper off at first, but then flatten out. This is called a heavy tailed distribution and it tends to produce outliers, extreme values, on both sides. This is not a bell shaped curve.

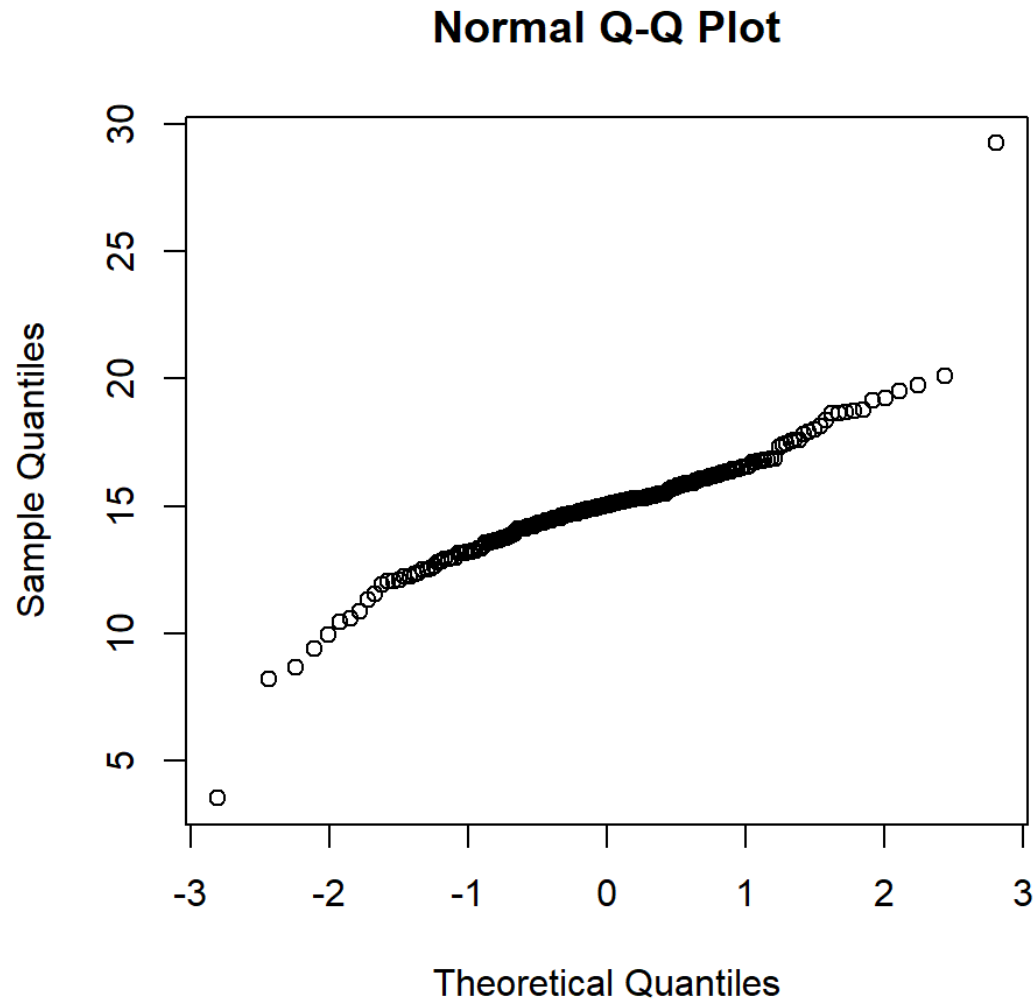
Heavy-tailed distribution, boxplot



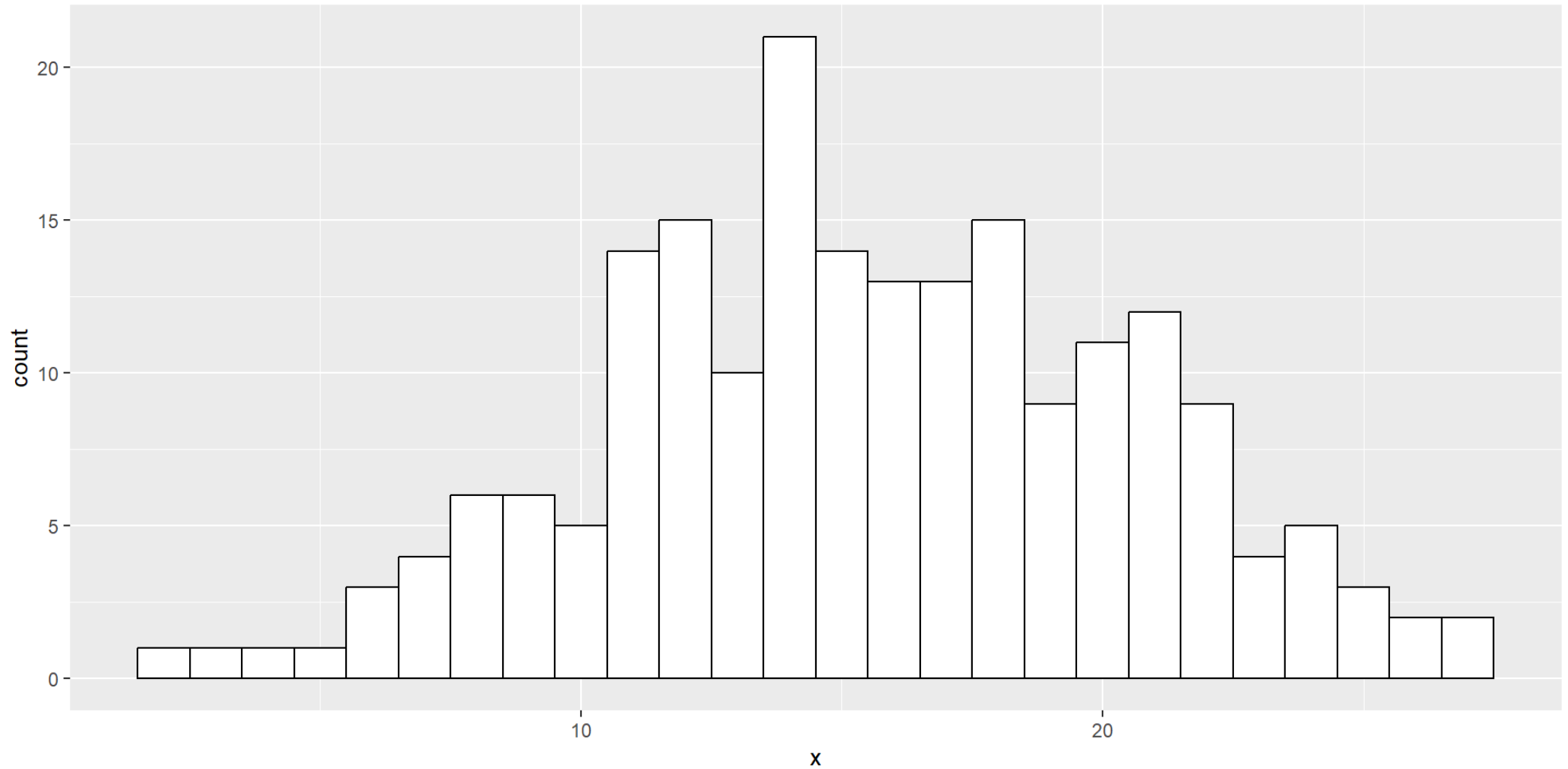
Speaker notes

The boxplot is not as useful as the histogram for detecting bimodal distributions.

Heavy tailed data, qq plot



A normal distribution, histogram



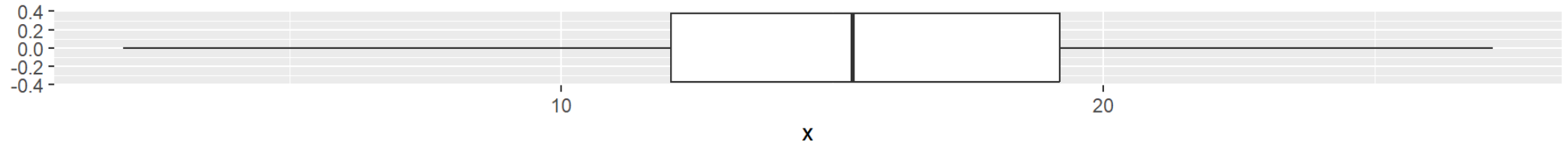
Speaker notes

Speaker notes

Here's a histogram that shows a symmetric distribution, with the most frequent values in the center and frequencies that taper off on either side.

This is a bell shaped curve.

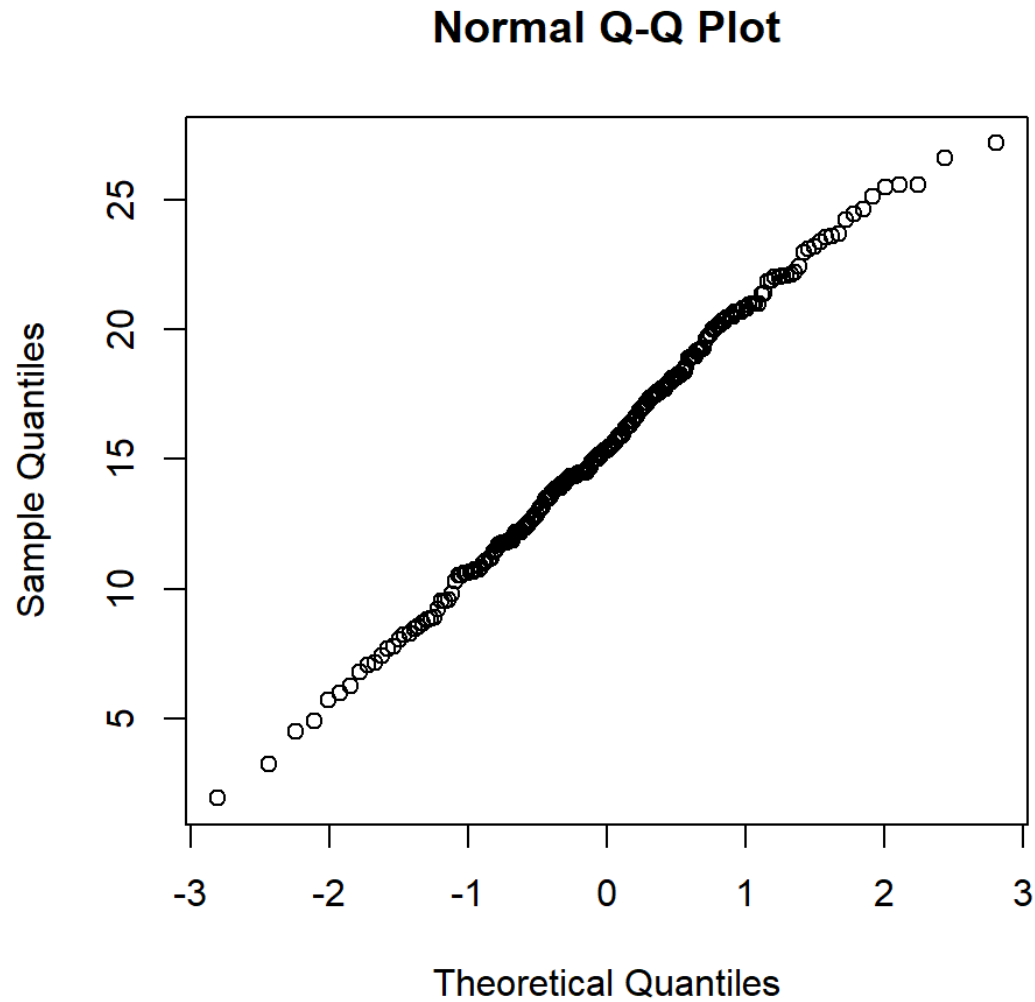
A normal distribution, boxplot



Speaker notes

The boxplot has a roughly symmetric box and roughly symmetric whiskers. The whiskers are a bit longer than the box itself, but not a lot longer.

A normal distribution, qq plot



Speaker notes

Speaker notes

A roughly straight line indicates a normal distribution.

Break #3

- What you have learned
 - Assessing normality
- What's coming next
 - Using R to assess normality

Live demo

Refer to

- The data set, [fev.csv](#)
- Its data dictionary, [fev.yaml](#)
- The demo program, [simon-5501-03-demo.qmd](#)

Break #4

- What you have learned
 - Using R to assess normality
- What's coming next
 - Your homework

Summary

- What you have learned
 - The normal distribution
 - Normal probabilities and quantiles
 - Assessing normality
 - Using R to assess normality
 - Your homework